Chong Duan

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Executive Summary

AI scientific leader with 10+ years of experience translating advances in multimodal AI, computational imaging, and quantitative biomarkers into clinical development and precision medicine. Proven track record defining AI strategy, leading enterprise-scale initiatives, and delivering regulatory-grade AI solutions spanning biomarker discovery, patient stratification, outcome prediction, and clinical trial innovation across oncology and other therapeutic areas.

Experience

Pfizer – Cambridge, MA

Director, Digital & Imaging AI Lead	**Translational Clinical Sciences**	04/2023 – present
Assoc. Director	--	07/2021 – 04/2023
Manager	--	07/2019 – 07/2021

1. Defined scientific strategy and led the development and deployment of AI-enabled biomarkers, computational diagnostics, multimodal models, and digital endpoints across Pfizer's clinical development portfolio to support dose optimization, patient stratification, early efficacy assessment, and clinical development decision-making.
2. Established **prospective validation** strategies for AI-assisted tumor response endpoints aligned with FDA CPIM recommendations, generating regulatory-grade evidence while navigating operational, regulatory, legal, and data privacy challenges.
3. Selected scientific contributions:
    a. Deep learning-based prognostic and predictive biomarkers - **ASCO 2024**, **JCO CCI 2024** [Invited paper], **ASCO 2025**.
    b. AI-powered RECIST v1.1 and beyond in clinical trials - **ESMO 2023**, **ESMO 2025** [**ESMO Daily Reporter Highlight**].
    c. Multimodal learning (clinical imaging + ctDNA) for early on-treatment prediction of long-term survival outcomes - **WCLC 2026.**
    d. Prospective validation of AI-assisted RECIST 1.1 in a phase 1 study of fetrastobart vedotin in patients with NSCLC - **ESMO 2026**.
4. Led multidisciplinary teams comprising direct reports and matrixed contributors across AI, clinical development, operations, biometrics, imaging sciences, regulatory, and external collaborators across academia, AI companies, and public-private partnerships.
5. Led scientific evaluation, benchmarking, and deployment of foundation models (Vision & Multimodal), establishing scalable governance for secure and compliant AI adoption across clinical development.
6. Leveraged cloud platforms for the development and deployment of advanced deep learning models:
    a. Fully automated echocardiogram analysis framework on AWS that reduces analysis time by >92% - **AWS Architecture Blog**, **Paper**, **GitHub**. [Pfizer Breakthrough Science & Innovation Award - the highest honor for a Pfizer R&D colleague]
7. Represented Pfizer in strategic public-private collaborations advancing the validation, standardization, and regulatory acceptance of AI-enabled biomarkers and clinical endpoints:
    a. Friends of Cancer Research: **ai.RECIST**
    b. Innovative Health Initiative (IHI) consortium: **SYNTHIA**
    c. IHI consortium: **RECIST 2.0**
    d. FNIH Biomarkers Consortium: **Mucosal Healing in Ulcerative Colitis**

Invicro (now Perceptive) **– Boston, MA**

Study Director, Discovery Research 02/2019 – 07/2019

1. Led imaging studies supporting PK/PD, bio-distribution, toxicology, and translational research for Pharmaceutical and Biotech clients.

Harvard Medical School – Boston, MA

Research Fellow, Cardiovascular Imaging 09/2017 – 01/2019

1. Developed deep learning methods for accelerated MRI reconstruction (>4x scan acceleration).
2. Developed a novel non-contrast MRI method for myocardial scar detection (>50% increase in contrast-to-noise ratio). [**Code** & **MRM 2018**].

Schlumberger – Cambridge, MA

Research Intern, Sensor Physics 05/2016 – 10/2016

1. Developed a simulation tool for virtual oil-well logging apparatus prototyping [**JMR 2017**].

Washington University School of Medicine – St. Louis, MO

Graduate Research Assistant 08/2014 – 08/2017

1. Developed Bayesian and pharmacokinetic modeling approaches for cancer imaging data, resulting in multiple first author publications [**MRM 2016**, **MIB 2018**].

Education

MS in Computer Science (specialization in Machine Learning)
Georgia Institute of Technology 2020

PhD in Physical Chemistry (focusing on Cancer Imaging)
Washington University in St. Louis 2017

BS in Chemistry
Nankai University 2012

Technical Skills

Machine Learning: Multimodal AI, Foundation Models, Representation Learning, Survival Modeling, Outcome Prediction, Vision Transformers, CNN, RNN, VAE, GANs, traditional ML models

Platform: Python, PyTorch, TensorFlow, R, MATLAB, C++, Fortran, MONAI, OpenCV, scikit-learn, scikit-image, Pyradiomics, AWS, Azure, Git/GitHub, Hugging Face, Docker, Jupyter Notebook

Leadership & Awards

Breakthrough Science & Innovation Award (Pfizer)	2021
William E. Upjohn Prizes Award (Pfizer)	2020
Journal of Cardiovascular Magnetic Resonance (JCMR) Gold Star Reviewer Award	2019 & 2020
Dean's Dissertation Fellowship (Washington University in St. Louis)	2017
The ISMRM Summa Cum Laude Merit Award *(Top 5% oral presentations)*	2016 & 2017

Teaching Assistant Award (Washington University in St. Louis)	2014
Chemical Safety Contest winner (Washington University in St. Louis)	2012
First Prize of Excellence Scholarship (Nankai University)	2009 – 2011

Invited Talks & Oral Presentations

1. Joint Statistical Meeting (JSM) 2022, "Deep Learning for Image Analysis - Initial Development of Digital Diagnostic/Prognostic Algorithms: Analogies and Lessons Learned from Drug Development", Washington, DC, August 2022.
2. AWS Healthcare & Life Sciences Symposium 2022, "Developing and Deploying Deep Learning-based Echocardiography Analysis with AWS", March 2022.
3. 77th AALAS National Meeting, "Fully Automated Mouse Echocardiography Analysis with Deep Convolutional Neural Networks", October 2021.
4. Summit for Clinical Ops Executives (SCOPE) 2021 – "Clinical Image Management and Exploration at Pfizer", June 2020.
5. 22nd annual meeting of the Society for Cardiovascular Magnetic Resonance, "Non-contrast Myocardial Viability Assessment Using a Hybrid Native T1 and Magnetization Transfer Imaging Sequence", Bellevue, Washington, June 2019.
6. 27th annual meeting of the International Society of Magnetic Resonance in Medicine, "CMR in Kidney Failure: Non-Contrast Imaging", Montreal, Québec, Canada, 2019.
7. 22nd annual meeting of the Society for Cardiovascular Magnetic Resonance, "Focus Session 6: MR Techniques and Methods: Non-contrast CMR and New Contrast Mechanisms", Bellevue, Washington, 2019.
8. 25th annual meeting of the International Society of Magnetic Resonance in Medicine, "Irradiated Brain Parenchyma Promotes Virulent Proliferation of Naive Glioma Cells: Mouse Model of Recurrent Glioblastoma", Honolulu, Hawaii, May 2017.
9. 24th annual meeting of the International Society of Magnetic Resonance in Medicine, "Can anti-VEGF Antibody Reverse Radiation Necrosis? A Preclinical Investigation", Singapore, May 2016.

Selected Publications

Full list available on **Google Scholar**; ***Corresponding Author**.

1. Lu, S. L., Chang, Y. C., Liang, C. H., Chiang, P. L., Maresca, K., Pollom, E., Wilner K., Lu, J. T., Toffalorio, F., Giaccone G., and **Duan, C***. (2026). AI-Enabled Early Intracranial Volumetric Response Predicts Systemic Progression Free Survival in the Phase III CROWN Study. *Neuro-Oncology Advances, 8(1), vdag236.*
2. Zhang, H., Wang, I. M., Rifi, N., Lu, J. T., Deng, S., Solomon, B., and **Duan, C.*** (2026) Early on-treatment prediction of long-term PFS using a joint AI-imaging and ctDNA model in the CROWN study. *Journal of Thoracic Oncology 21.10.*
3. Liu, A., **Duan, C.,** Maresca, K., Pithavala, Y., Lu, J. T., Guo, C., and Hibma, J. (2026). AI-powered Lesion-level Tumor Growth Inhibition Modeling Improves Model Stability and Prognostic Association with PFS. CPT: Pharmacometrics & Systems Pharmacology 15.6: e70279.
4. Lu, S. L., Chang, Y. C., Liang, C. H., Chiang, P. L., Maresca, K., Pollom, E., ... & **Duan, C*.** (2025). 2012P A post-hoc analysis of the CROWN study in ALK+ NSCLC using AI to predict progression-free survival based on early response. *Annals of Oncology*, 36, S1146.
5. Sako, C., Schmidt, T., **Duan, C.,** Maresca, K., Gowan, A. C., Vyas, S., ... & Kelly, R. J. (2025). Association of deep learning CT response assessment and interpretable components with overall survival in advanced NSCLC: Validation in a trial of sasanlimab and a real-world dataset. *J Clin Oncol **43**, 1559.*

6. Sako, C., **Duan, C.**, Maresca, K., Kent, S., Schmidt, T.G., Aerts, H.J., Parikh, R.B., Simon, G.R. and Jordan, P., 2024. Real-World and Clinical Trial Validation of a Deep Learning Radiomic Biomarker for PD-(L) 1 Immune Checkpoint Inhibitor Response in Advanced Non–Small Cell Lung Cancer. *JCO Clinical Cancer Informatics*, *8*, p.e2400133.
7. Montgomery, M.K., **Duan, C.**, Mansukh, L., Chang, S., Cubias, A., Brun, S., Giddabasappa, A. and Jiang, Z.K., (2024), Applying deep learning to segmentation of murine lung tumors in pre-clinical micro-computed tomography. *Translational Oncology*, *40*, p.101833.
8. Lu, S.-L., **Duan, C.**, Chang, Y.-C., Liang, C.-H., Chiang, P.-L., Lin, V., Yang, Y.-S., Schayowitz, A., Selaru, P., Wilner, K. D., & Lu, J.-T. (2023), 1421P AI-powered intracranial tumor response predicts systemic progression with high concordance in endpoint evaluation in the phase III CROWN study. *Annals of Oncology*, 34, S811. https://doi.org/10.1016/J.ANNONC.2023.09.2452.
9. **Duan C**, Montgomery MK, Chen X, Ullas S, Stansfield J, McElhanon K and Hirenallur-Shanthappa D, 2022. "Fully automated mouse echocardiography analysis using deep convolutional neural networks," *American Journal of Physiology-Heart and Circulatory Physiology*, 323(4), pp.H628-H639.
10. Zhu Y, Fahmy AS, **Duan C**, Nakamori S, Nezafat R. Automated Myocardial T2 and Extracellular Volume Quantification in Cardiac MRI Using Transfer Learning–based Myocardium Segmentation. *Radiology: Artificial Intelligence (2020)*, 2(1), e190034.
11. **Duan C**, Yang R, Yuan L, et al. Late effects of radiation prime the brain microenvironment for accelerated tumor growth. *Int J Radiat Oncol Biol Phys* 103.1 (2019): 190-194.
12. **Duan C**, Zhu Y, Jang J, et al. Non-Contrast Assessment of Myocardial Viability using a Hybrid Native T1 and Magnetization Transfer Imaging Sequence. *Magn Reson Med.* (2018) https://doi.org/10.1002/mrm.27636
13. **Duan C**, Kallehauge JF, Pérez-Torres CJ, et al. Modeling Dynamic Contrast-Enhanced MRI data with a Constrained Local AIF. *Mol Imaging Biol* (2018) 20: 150. https://doi.org/10.1007/s11307-017-1090-x.
14. **Duan C**, Kallehauge JF, Bretthorst GL, et al. Are Complex DCE-MRI Models Supported by Clinical Data? *Magn Reson Med* (2017) 77:1329-1339.
15. **Duan C**, Ryan C, Utsuzawa S, et al. Effect of Off-Resonance on T_1 Saturation Recovery Measurement in Inhomogeneous Field. J Magn Reson (2017) 281: 31-43.
16. **Duan C,** Pérez-Torres CJ, Yuan L, et al. Can anti-Vascular Endothelial Growth Factor Antibody Reverse Radiation Necrosis? A Preclinical Investigation (**Cover Article**). *J Neurooncol* (2017) 133: 9-16.